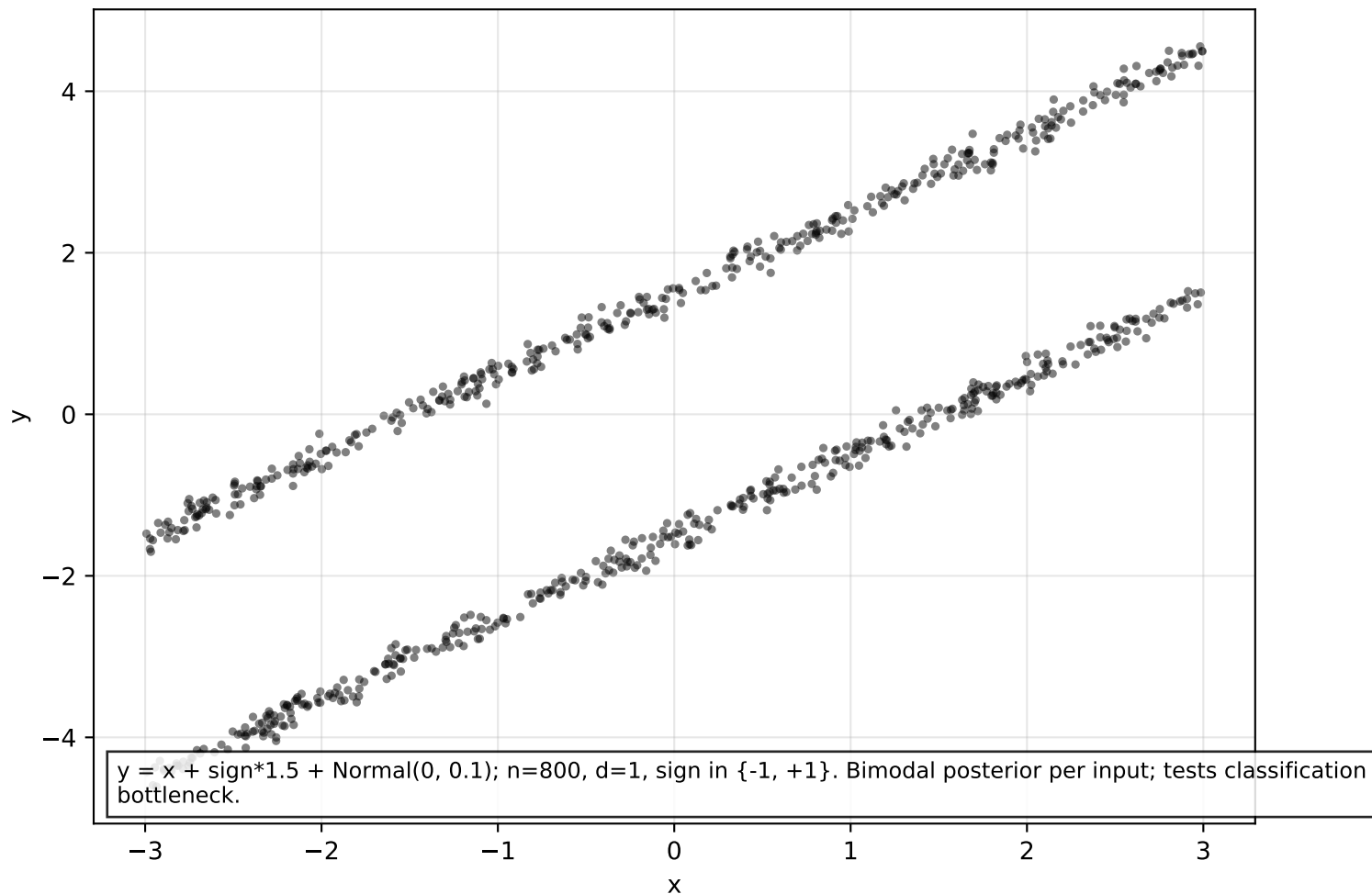


Dataset: bimodal (n=800)

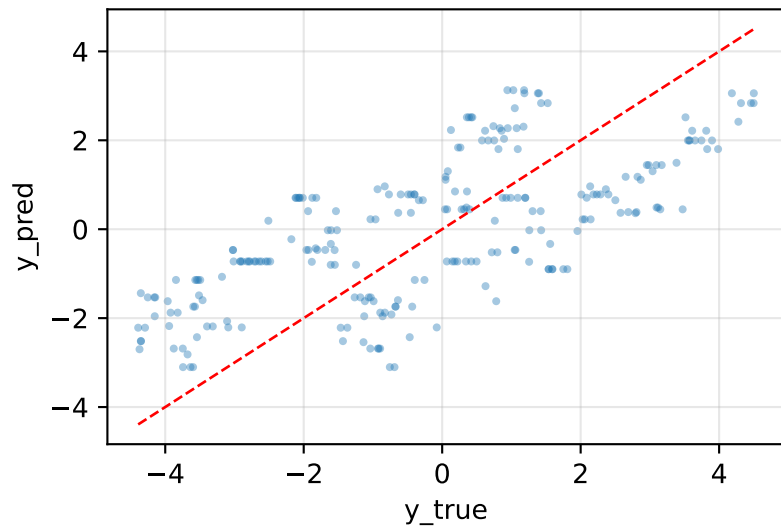


Model comparison

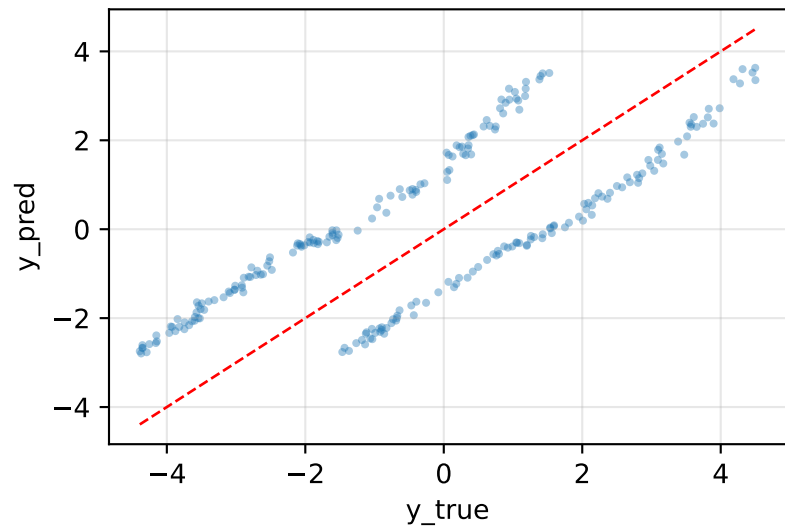
model	mse	nll	crps	picp95	mpiw95	sharpness
XGBoost	2.8745	2.0522	1.0461	0.9000	4.9584	1.2649
NN(PROB)	2.3927	1.8558	0.9299	1.0000	6.0127	1.5339
ProbReg(k=3)	2.3625	1.8499	0.9251	1.0000	5.9271	1.5120
ClassReg(k=7)	2.8426	1.9904	0.9912	1.0000	8.2123	2.0950
FlexNN(ELBO)	2.5661	1.9138	0.9679	1.0000	5.9333	1.5136

Predicted vs actual — bimodal

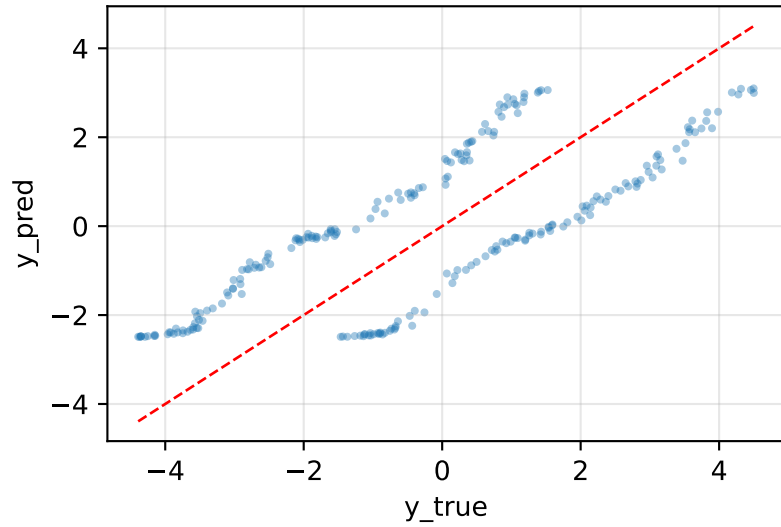
XGBoost MSE=2.874



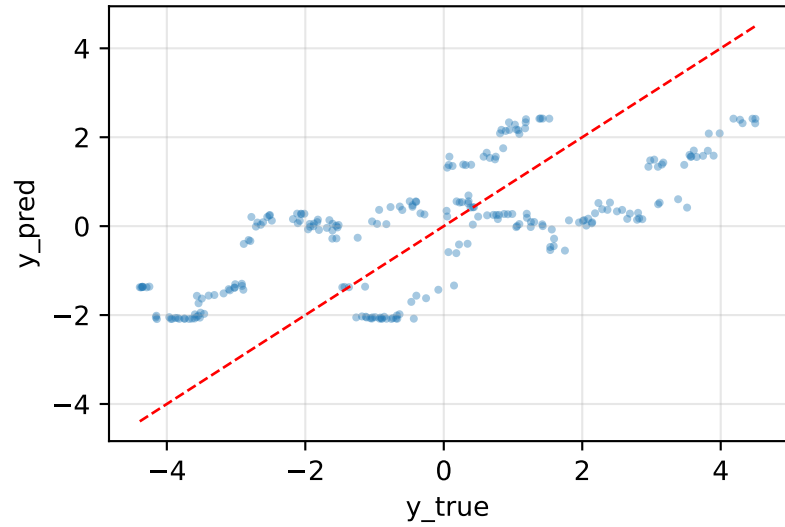
NN(PROB) MSE=2.393



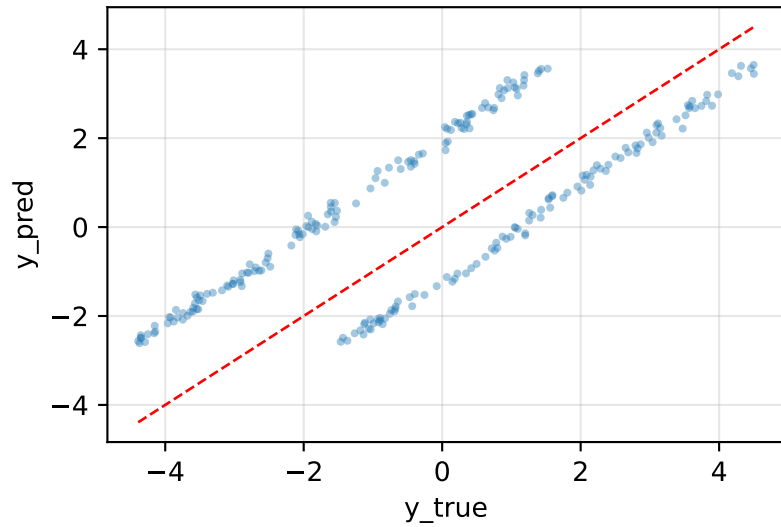
ProbReg(k=3,SEP) MSE=2.363



ClassReg(k=7) MSE=2.843

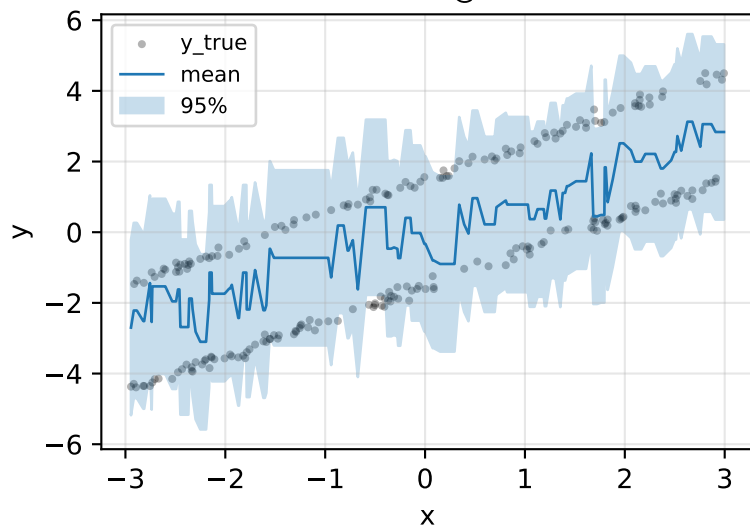


FlexNN(ELBO,PROB) MSE=2.566

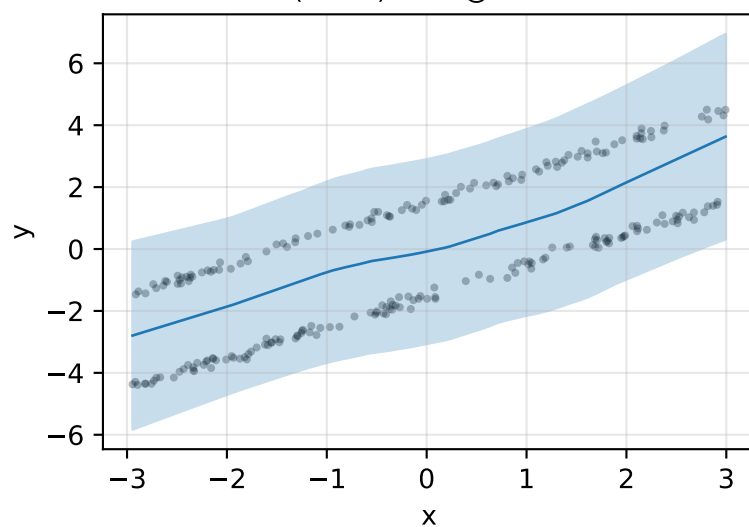


Prediction intervals — bimodal

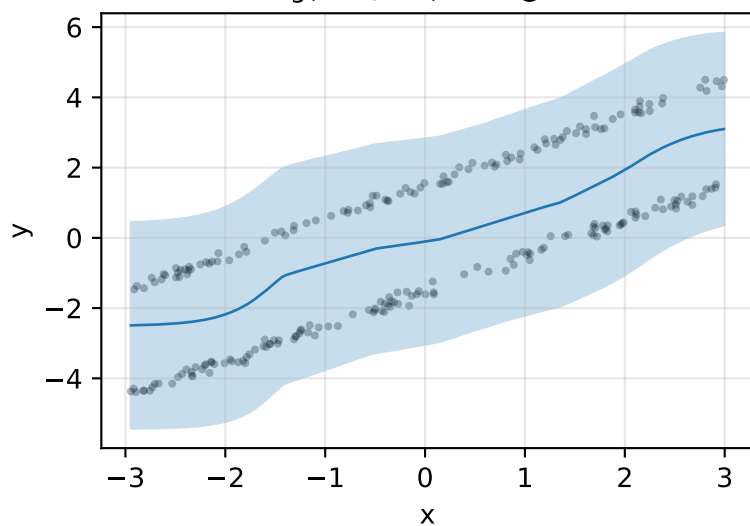
XGBoost PICP@95=0.90



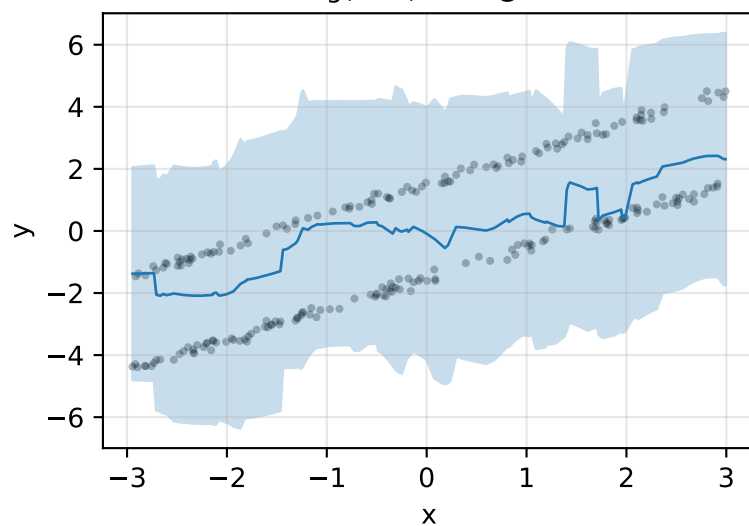
NN(PROB) PICP@95=1.00



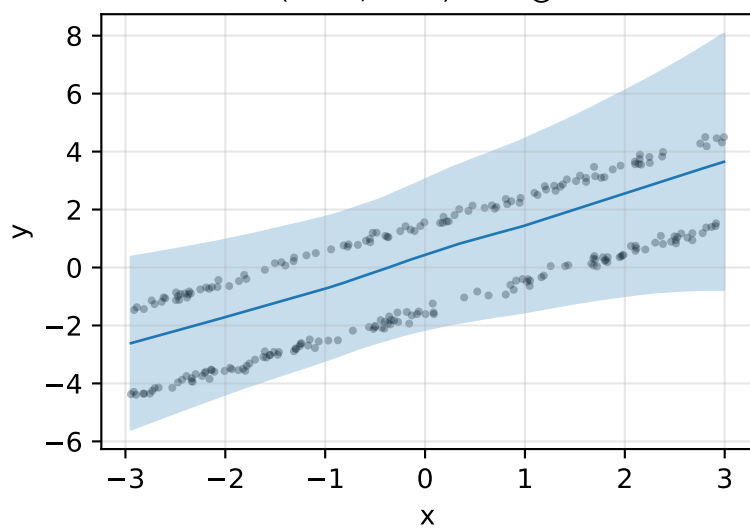
ProbReg(k=3,SEP) PICP@95=1.00



ClassReg(k=7) PICP@95=1.00

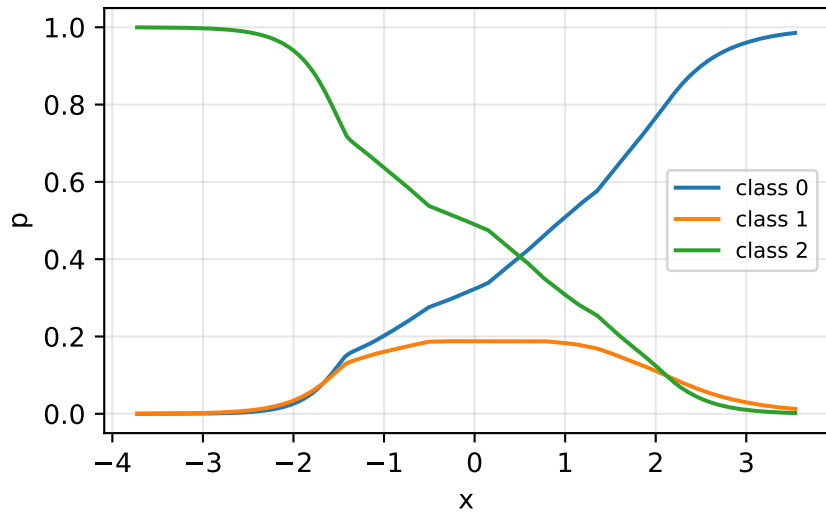


FlexNN(ELBO,PROB) PICP@95=1.00

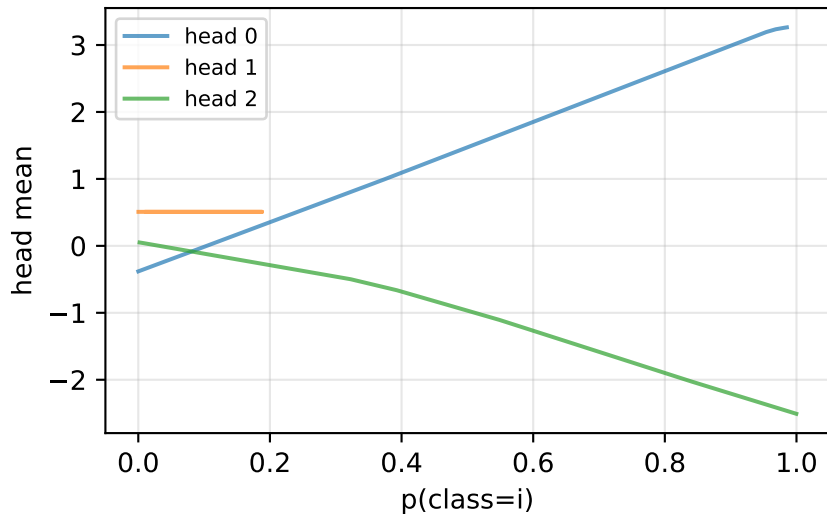


ProbReg internal probabilities + head outputs

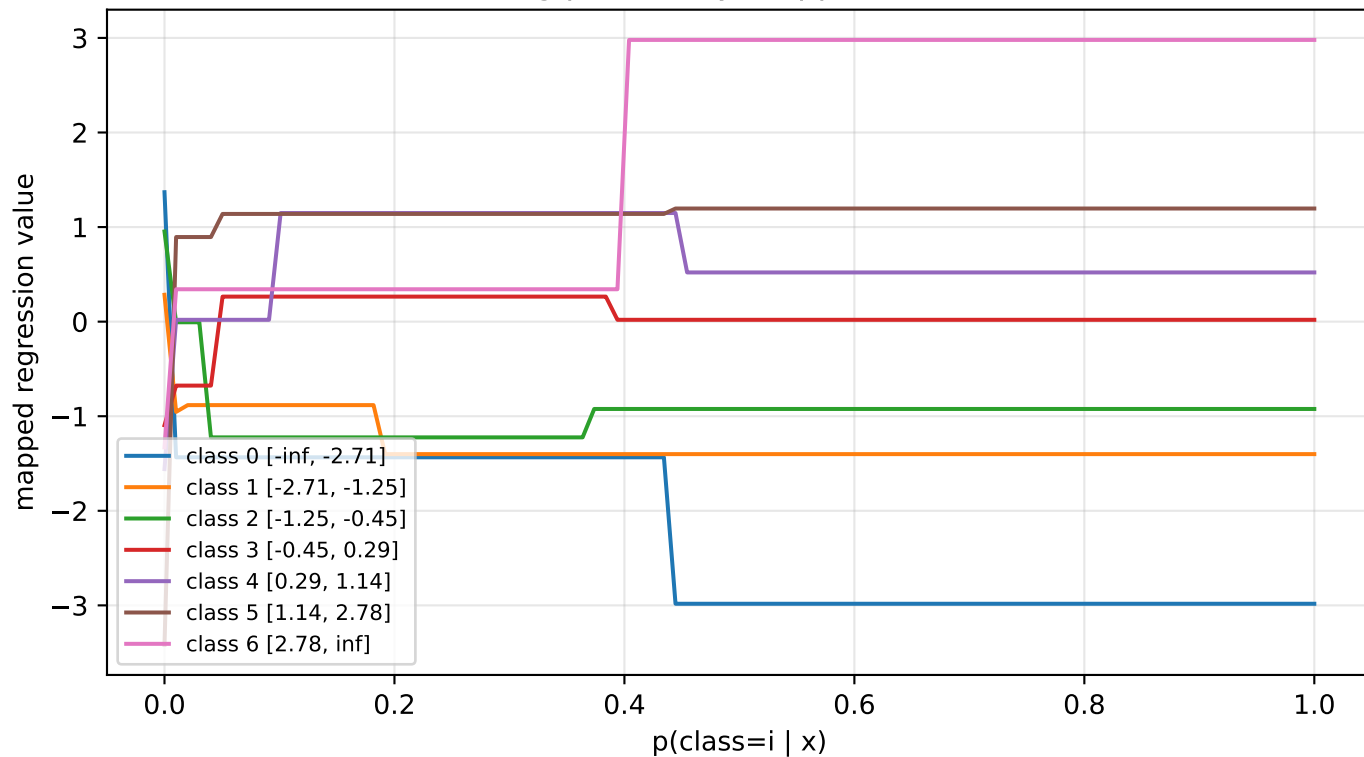
Classifier $p(\text{class} \mid x)$



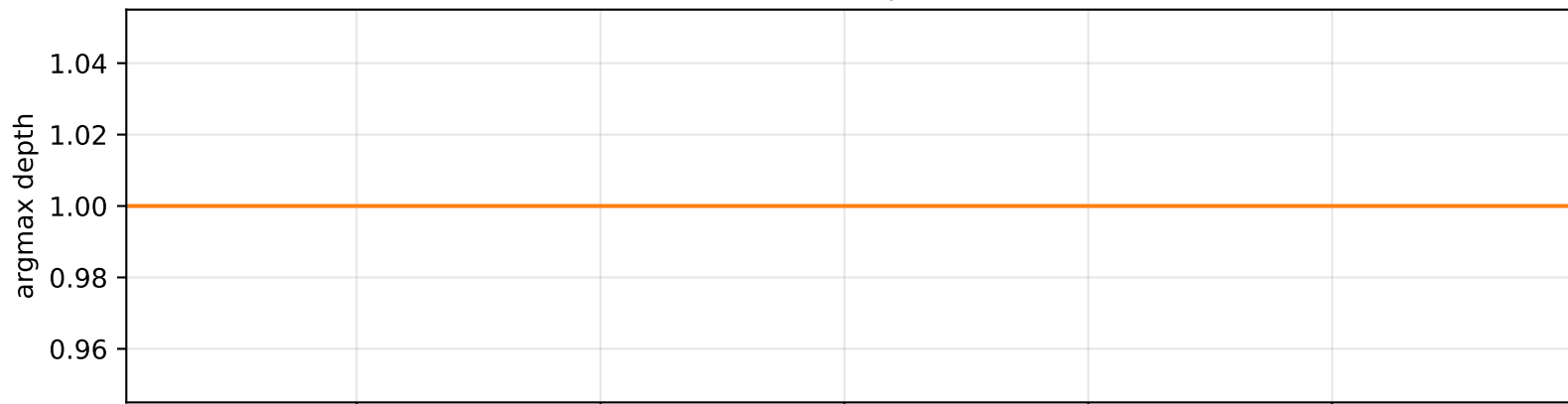
Regression head mean vs class prob



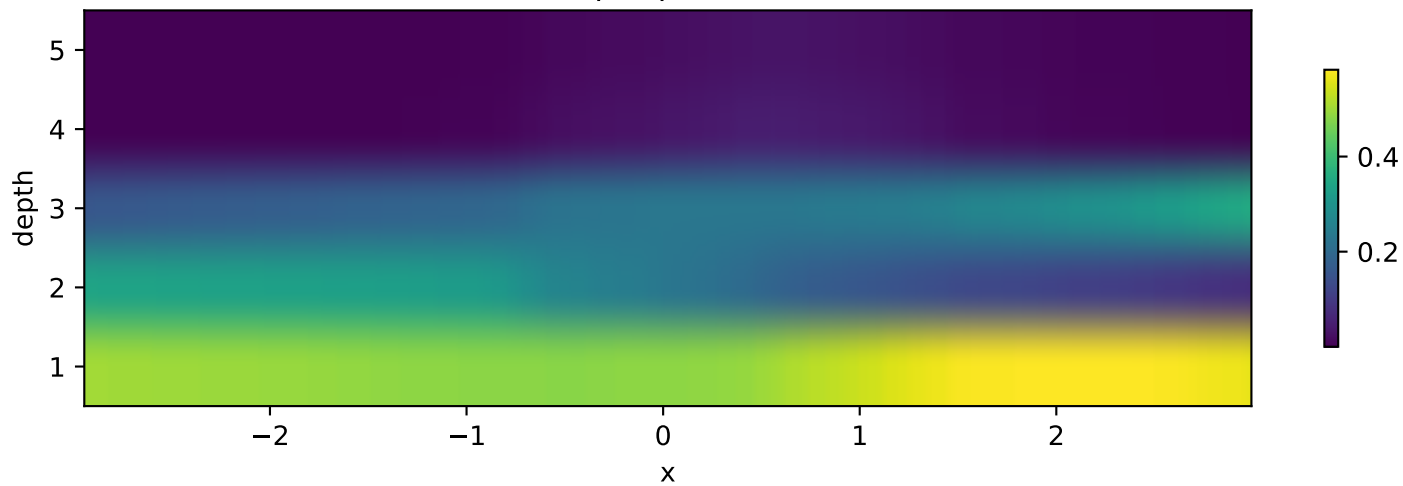
ClassReg probability mappers — bimodal



FlexNN selected depth — bimodal



Soft depth probabilities



Ranked MSE for bimodal:

- ProbReg(k=3,SEP): MSE=2.3625 NLL=1.850 PICIP95=1.00
- NN(PROB): MSE=2.3927 NLL=1.856 PICIP95=1.00
- FlexNN(ELB0,PROB): MSE=2.5661 NLL=1.914 PICIP95=1.00
- ClassReg(k=7): MSE=2.8426 NLL=1.990 PICIP95=1.00
- XGBoost: MSE=2.8745 NLL=2.052 PICIP95=0.90

Notes:

Bimodal: $y = x \pm 1.5$ has a genuinely two-modal posterior. ClassReg (k even) or ProbReg SEP (k=2